

DATA 621 Blog 3

Cross Validation R examples (caret)

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Cross-Validation

In statistical modeling and machine learning, evaluating how well a model generalizes to unseen data is just as important as achieving strong performance on the training dataset. A model that fits the training data extremely well may fail to perform adequately on new observations, a problem commonly known as overfitting. Cross-validation provides a systematic framework for estimating out-of-sample performance and reducing this risk.

Cross-validation works by repeatedly splitting the available data into training and validation subsets. The model is trained on one portion of the data and evaluated on the remaining portion, allowing performance to be assessed on data not used during model fitting. By repeating this process across multiple splits, cross-validation produces a more reliable estimate of a model's predictive ability than a single train-test split.

One of the most widely used approaches is k -fold cross-validation, where the dataset is divided into k roughly equal folds. The model is trained k times, each time leaving out one fold for validation and using the remaining $k - 1$ folds for training. Performance metrics are then averaged across all folds, reducing variability due to random data partitioning.

Cross-validation is especially valuable when comparing multiple models or tuning hyperparameters, as it allows competing approaches to be evaluated under consistent conditions. By focusing on generalization rather than in-sample fit, cross-validation helps ensure that selected models are robust, interpretable, and better suited for real-world deployment.

A helpful way to think about cross-validation is through the analogy of a student preparing for a final exam. If the student only practices with one set of sample questions, they may

simply memorize those questions rather than truly understand the material, an example of overfitting. Cross-validation is like having multiple practice exams: the student studies from most of them and tests themselves on the remaining one, rotating through all sets. By the end, the student has been evaluated on all material, ensuring genuine understanding rather than success or failure due to familiarity with a single set of questions.

Caret

The **caret** package is a powerful and widely used tool that provides a unified interface for implementing a broad range of statistical and machine learning algorithms. One of its key strengths is the ability to easily incorporate cross-validation and other resampling techniques into modeling workflows through a consistent and flexible framework. By standardizing model training, tuning, and evaluation, **caret** simplifies the process of comparing models and assessing their generalization performance.

```
# Define cross-validation method
# number is k parameter or folds
cv_control <- trainControl(method = "cv", number = 5)

# Train model with 5-fold CV passing on training control method, to the model function
cv_model_lm <- train(
  mpg ~ wt,
  data = mtcars,
  method = "lm",
  trControl = cv_control
)
# summarize the results
summary(cv_model_lm)
```

Call:

```
lm(formula = .outcome ~ ., data = dat)
```

Residuals:

Min	1Q	Median	3Q	Max
-4.5432	-2.3647	-0.1252	1.4096	6.8727

Coefficients:

```

              Estimate Std. Error t value Pr(>|t|)
(Intercept)  37.2851      1.8776  19.858 < 2e-16 ***
wt           -5.3445      0.5591  -9.559 1.29e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.046 on 30 degrees of freedom
Multiple R-squared:  0.7528,    Adjusted R-squared:  0.7446
F-statistic: 91.38 on 1 and 30 DF,  p-value: 1.294e-10

```

```
cv_model_lm
```

Linear Regression

32 samples
1 predictor

No pre-processing
Resampling: Cross-Validated (5 fold)
Summary of sample sizes: 26, 25, 25, 26, 26
Resampling results:

RMSE	Rsquared	MAE
3.080738	0.7978128	2.50402

Tuning parameter 'intercept' was held constant at a value of TRUE

Regression

The `cv_model` output summarizes the results of fitting a linear regression model using 5-fold cross-validation. It reports performance metrics such as RMSE, R^2 , and MAE averaged across all folds, providing an estimate of how well the model is expected to perform on unseen data. The resampling information confirms that the dataset was repeatedly partitioned into training and validation sets, ensuring that model evaluation is not dependent on a single split.

The `summary(cv_model)` output presents the final fitted model using the full dataset, including coefficient estimates, standard errors, and significance tests. While this summary reflects the in-sample fit, the cross-validated metrics from `cv_model` offer a more reliable assessment of generalization performance. Together, these outputs allow both model interpretation and robust evaluation of predictive accuracy.

The difference in R^2 values arises because they measure performance under different conditions. The R^2 reported in `summary(cv_model)` reflects **in-sample fit**, indicating how well the model

explains variability in the data it was trained on. In contrast, the cross-validated R^2 reported by `cv_model` is an **out-of-sample estimate**, averaged across validation folds, and is typically lower because it evaluates the model on unseen data. This gap is expected and highlights the importance of cross-validation in providing a more realistic measure of generalization performance.

In this case, the cross-validated R^2 (0.79) is slightly higher than the in-sample R^2 from the final model (0.75). This can occur due to **sampling variability**, especially with a relatively small dataset, where different training folds may better capture the underlying linear relationship than the full dataset does on average. Because cross-validated R^2 is computed as an average across multiple validation splits, it can occasionally exceed the single in-sample estimate without indicating overfitting or model error.

Classification

Cross validation can be used a variety of classifications provided within the `caret` package.

```
data(iris)

# Define 5-fold cross-validation
cv_control <- trainControl(
  method = "cv",
  number = 5
)

# Train classification model
cv_model_cat <- train(
  Species ~ Petal.Length + Petal.Width,
  data = iris,
  method = "rf",
  trControl = cv_control
)
```

note: only 1 unique complexity parameters in default grid. Truncating the grid to 1 .

```
cv_model_cat
```

Random Forest

```
150 samples
 2 predictor
 3 classes: 'setosa', 'versicolor', 'virginica'
```

No pre-processing
Resampling: Cross-Validated (5 fold)
Summary of sample sizes: 120, 120, 120, 120, 120
Resampling results:

Accuracy	Kappa
0.96	0.94

Tuning parameter 'mtry' was held constant at a value of 2

```
cv_model_cat$finalModel
```

Call:

```
randomForest(x = x, y = y, mtry = param$mtry)
      Type of random forest: classification
      Number of trees: 500
No. of variables tried at each split: 2
```

OOB estimate of error rate: 4.67%

Confusion matrix:

	setosa	versicolor	virginica	class.error
setosa	50	0	0	0.00
versicolor	0	47	3	0.06
virginica	0	4	46	0.08

Calling the resampled model object returns aggregated performance metrics averaged across all trained models from the cross-validation process. These results provide insight into overall model performance, while the final fitted model can be examined separately for coefficient estimates and interpretation. The [caret package](#) offers thorough documentation that explains how to access and interpret the various components stored within a trained model object.

Conclusion

Cross-validation is an essential technique for assessing how well a model generalizes beyond the training data and for guarding against overfitting. By evaluating performance across multiple resampled subsets, it provides a more reliable estimate of out-of-sample behavior than a single train-test split.

The **caret** package makes implementing cross-validation straightforward by offering a consistent and intuitive interface across many modeling approaches. This combination of methodological rigor and practical simplicity makes cross-validation with **caret** a valuable tool in modern statistical modeling workflows in R.